

Abish Bhusal

Backend Software Engineer (Go, Python, C++)

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WORK EXPERIENCE

• Nube iO

Remote

Software Engineer (C++, Go)

Nov 2024 – Present

Distributed Automation Architecture (C++ / Go / NATS):

- Developing a high-performance, distributed automation engine using C++ and Go, leveraging NATS for real-time messaging and Cap'n Proto for high-efficiency serialization.
- Engineered a modular extension system enabling the engine to scale across diverse industrial protocols and high-throughput data environments.
- Architected a thread-safe, priority-based polling engine with starvation prevention, supporting deterministic data collection from thousands of concurrent field devices.
- Built a high-performance Modbus TCP/RTU extension in C++ featuring multi-register decoding and real-time state synchronization via deterministic poll-cycles.
- Designed atomic bulk-operation APIs and a multi-threaded database writer to ensure zero-latency engine performance under heavy industrial loads.

• Rara Labs Private Limited

Lalitpur, Nepal

Backend Developer

Apr 2022 – Oct 2024

System Architecture & Multi-Tenancy (Go):

- Engineered and maintained a layered backend in Go, enforcing strict separation of concerns to support a scalable, feature-rich accounting system called Tigg.
- Implemented and upheld the core multi-tenancy architecture (Tenant wise database).
- Designed the asynchronous job processing system for handling resource-intensive background tasks like financial report generation and data synchronization.
- Designed and secured public facing APIs for core modules like accounting, reporting with signature verification, rate limiting, and Redis-based nonce tracking.

Database Engineering & Administration (PostgreSQL):

- Authored and optimized complex SQL queries for the platform's core modules, including accounting, inventory, and reporting, leveraging advanced PostgreSQL features such as CTEs, window functions, and custom aggregates.
- Tuned performance by analyzing query plans (EXPLAIN ANALYZE) and adding indexes to significantly reduce API latency for heavy endpoints.
- Managed database connection pooling to reliably support thousands of concurrent tenant connections.
- Built and maintained a zero-downtime schema migration tool, enabling safe, version-controlled database updates.
- Designed and automated backup and point-in-time recovery strategies to ensure data integrity and disaster readiness.

SKILLS, LANGUAGES, INTERESTS

- **Languages:** English (Professional working proficiency), Nepali (Native proficiency)
- **Programming:** Golang, PostgreSQL, C/C++, Python, Lua, NATS
- **Software Engineering:** Software Infrastructure, Industrial IoT (Modbus), Database Administration, Concurrent Programming & Thread Safety
- **Interests:** Video Games, Football, Reading superhero comics

EDUCATION

IOE, Pulchowk Campus , Tribhuvan University

Pulchowk, Lalitpur, Nepal

Bachelors in Computer Engineering

Aug 2018 – Jul 2023

- **Courses:** Operating Systems, Theory of Computation, Data Structures and Algorithms, C, OOP using C++, Artificial Intelligence and Machine Learning, Networking, Databases
- **Major Project:** Developed a Nepali OCR system using Convolutional Recurrent Neural Networks (CRNN) and Long Short-Term Memory (LSTM) networks, achieving 86% accuracy on a custom dataset of handwritten Nepali characters.
- **Extra Curricular:** Python instructor in a software fellowship organized by LOCUS , Organized a competitive programming hackathon called Yomari CodeCamp, Participated in multiple hackathons in my junior years

St. Xavier's College

Maitighar , Kathmandu, Nepal

High School major in Science

Aug 2016 – Jul 2018

- Graduated with Distinction, scoring 3.6/4.0 GPA in Science stream with Physics, Chemistry, and Maths.

CERTIFICATIONS

AWS Academy graduate-AWS Academy Cloud Foundations

Issued by *AWS Academy*

AWS Academy

Sep 2022

AWS Academy graduate-AWS Academy Cloud Architecting

Issued by *AWS Academy*

AWS Academy

Aug 2021

Winter school in AI

NAAMI Nepal

RESEARCH AND PUBLICATIONS

CRNN based Nepali OCR

Published on *Tribhuvan University*

Feb 2023